

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

VI Semester of B.Sc. (Biotechnology) Examination April-May, 2019

Course code & name - BT603: Basic Animal Biotechnology

**Date: 02/05/2019      Day: Thursday      Time: 1:30 PM to 1:50 PM      Maximum Marks: 10**

## MCQ

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### Important Instructions:

- Tick (✓) the correct answer and answer all the questions in the question paper itself.
  - Use of non-programmable calculator is allowed.
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**Q. 1    Choose the correct answer for the following questions.**

**10**

1. Which of the following is a cryoprotective agent used in cryopreservation of mammalian cells?

- |                        |                              |
|------------------------|------------------------------|
| a. Acetic acid (AA)    | b. Dimethyl sulfoxide (DMSO) |
| c. Toluidine blue (TB) | d. Both 'a' and 'c'          |

2. Microinjection is a technique used for \_\_\_\_\_

- |                                   |                                 |
|-----------------------------------|---------------------------------|
| a. Transformation of animal cells | b. Transfection of animal cells |
| c. Transduction                   | d. None of the above            |

3. Which of the following is/are application/s of Animal Biotechnology?

- |                    |                                     |
|--------------------|-------------------------------------|
| a. Animal cloning  | b. Production of transgenic animals |
| c. Cancer research | d. All of the above                 |

4. What is the role of 5% CO<sub>2</sub> in gas phase, when the cells are grown in a CO<sub>2</sub> incubator?
- a. Prevents contamination
  - b. Buffering
  - c. Maintains atmospheric pressure
  - d. Required as carbon source
5. Which of the following is/are characteristic/s of transformed cells?
- a. They are immortal
  - b. Can grow only up to few generations
  - c. Slow growing
  - d. All of the above
6. What is the role of promoter sequence in transgene construct?
- a. Gene replication
  - b. Gene expression
  - c. Attachment of DNA polymerase
  - d. Both 'a' and 'c'
7. \_\_\_\_\_ is used for arresting the cells at metaphase while performing transfer of transgene into the animal cell.
- a. Colcemid
  - b. DEAE-dextran
  - c. Calcium phosphate
  - d. Lipopolysaccharide
8. HELA is a \_\_\_\_\_ cell line.
- a. Normal
  - b. Rapid
  - c. Continuous
  - d. Human kidney
9. What are multipotent stem cells?
- a. Can form all cell types in body
  - b. Can form all cell types in body and placental cells
  - c. Can form a specific few types of cells in body
  - d. None of the above
10. Liquid nitrogen is used for \_\_\_\_\_
- a. Transformation
  - b. Cell viability counts
  - c. Protection against freezing
  - d. Cryopreservation

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VI Semester of B.Sc. (Biotechnology) Examination April-May, 2019

Course code & name - BT603: Basic Animal Biotechnology

Date: 02/05/2019 Day: Thursday Time: 1:51 PM to 3:30 PM Maximum Marks: 25

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## Important Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
  2. Make suitable assumptions and draw neat figures wherever required.
  3. Use of non-programmable calculator is allowed.
  4. Show necessary calculations.
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## SECTION I

**Q. 2 Answer the following questions as directed. 10**

**Q. 2 A Answer ANY TWO of the following questions. 6**

1. Describe any three applications of transgenic animals with examples.
2. Enlist all the major instruments used in animal cell culture laboratory with their respective uses.
3. Enlist common contaminants in the cell culture. Name any four methods used for detecting contamination and discuss about PCR detection method.

**Q. 2 B Write a short note on ANY TWO of the following methods used for introduction of transgene into the animal cells. 4**

1. Electroporation
2. Gene gun
3. Calcium phosphate precipitation

## SECTION II

**Q. 3 Answer the following questions as directed. 15**

**Q. 3 A Answer ANY TWO of the following questions. 8**

1. What are continuous cell lines? Define, enlist the characteristics of transformed cells and give any two examples of continuous cell lines.
2. Describe (a) Artificial Insemination and (b) Multiple Ovulation and Embryo Transfer techniques used for livestock improvement.

3. Write a note on viral vectors used for introduction of transgene into the cells, give four examples of viral vectors and discuss the retrovirus-mediated gene transfer.

**Q. 3 B      Answer ANY TWO of the following questions.      4**

1. Write down any four advantages and disadvantages of serum-free media.
2. Enlist various methods used for screening the transgenic positives.
3. Enlist major functions of serum in cell culture media.

**Q. 3 C      Write a short note on ANY ONE of the following.      3**

1. Cryoprotective agents and types.
2. Substrates used for culturing animal cells.